diapers and beer, since it’s one of the most quoted data science anecdotes.

Figure 5.2 illustrates what we would like to achieve.

Initially, since we have no data at all, we consider every product as a group of its own, or if you prefer, no two products will have an association.

When customers buy two products together frequently, we then establish a link between them, considering both part of the same group – to keep things simple, imagine that the rule set by the data science team is that if it happens more than a fixed threshold during the last hour that item X and item Y are bought together, then their 2 categories are merged.

For instance, it might be: if during the last hour phones and tablets are bought together more than 500 times (or for more than 1% of total purchases), then they should be in the same group – so we merge their groups.

Then if a customer buys a product X, we can suggest them a random item from the same group.

This process described above is pretty common in data science: some of you might have recognized that it’s nothing less than hierarchical clustering. If that didn’t ring a bell, do not worry: we will expand on clustering in section 5.7.3.

This is, obviously, an extreme simplification. In a real non-personalized recommender system, we would track associations between products measuring the strength of the link as the confidence that when X is bought also Y is: for